

e-commerce-analysis (Short Version)

Prerequisites:

- Python 3.9+
- Docker & Docker Compose
- Git
- VS Code (recommended)

Quick Start

1. Clone the Repository

```
git clone https://github.com/jaybharuka/e-commerce-analysis.git
```

```
cd e-commerce-analysis
```

2. Create .env

```
DEBUG=True
```

```
STREAMLIT_THEME=dark
```

3. Install Dependencies

```
pip install -r requirements.txt
```

Run the Application

Option 1: Streamlit Dashboard

```
python -m streamlit run streamlit/app_complete.py --server.port=8502
```

Option 2: ML Models

```
python ml_analysis/run_all_analyses.py
```

Individual:

```
python ml_analysis/rfm_analysis.py
```

```
python ml_analysis/customer_segmentation.py
```

```
python ml_analysis/market_basket.py
```

```
python ml_analysis/sales_forecast_prophet.py
```

```
python ml_analysis/churn_prediction.py
```

```
python ml_analysis/product_recommendation.py
```

Option 3: Docker Deployment

```
docker-compose up -d
```

```
docker-compose logs -f streamlit
```

```
docker-compose down
```